8(a)(i)	Two bodies are in thermal equilibrium means they are at the same temperature.
8(a)(ii)	The rate of flow of thermal energy from one body to the other is exactly equal and opposite to the rate of flow of thermal energy in the opposite direction. Hence, there is no net movement of thermal energy between the two bodies.
8(b)(i)	The ice is not in thermal equilibrium with its surrounding because the ice is at a lower temperature than its surrounding. Hence, there is a net rate of thermal energy being transferred from its surrounding to the ice, melting the ice.
8(b)(ii)	Applying conservation of energy, Thermal energy gained by ice = Thermal supplied by electrical power + thermal energy gained from the surrounding $ml = Pt + H$ (where H is thermal energy gained from surrounding) $ml = VIt + H$ $H = ml - VIt$ $= (114.0 - 32.4)(330) - (12.0)(6.3)(5.0 \times 60)$ $= 4248$ J Rate of thermal energy gained from surrounding. $\frac{\Delta H}{\Delta t} = \frac{4248}{5.0 \times 60}$ $= 14.16$ ≈ 14.2 W
8(c)(i)	The internal energy of a system is the sum of the microscopic kinetic energy due to the random motion of its molecules and the microscopic potential energy due to its intermolecular forces of attraction. In an ideal gas, collisions are perfectly elastic and intermolecular forces of attraction are negligible. Hence, it has no microscopic potential energy and its internal energy consists purely of the microscopic kinetic energy. Temperature is a measure of the average kinetic energy of the molecules in a system. The average kinetic of a molecule is directly proportional to thermodynamic temperature, $\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$. Since the internal energy is the sum of all microscopic kinetic energy, it is also directly proportional to the thermodynamic temperature, $U = \frac{3}{2}NkT$.

8(c)(ii) 1.	Molar mass of neon-20 gas = 20 g mol^{-1} Using $pV = nRT$ $n = \frac{pV}{RT}$ In initial state, $n = \frac{1.0 \times 10^{5} \times 2.0 \times 10^{-2}}{8.31 \times (273.15 + 25)} = 0.80722$ mol Mass of gas, $m = 0.80722 \times 20 = 16.14 \approx 16$ g In the final state, $n = \frac{1.5 \times 10^{5} \times 2.0 \times 10^{-2}}{8.31 \times (273.15 + 174)} = 0.80736$ mol Mass of gas, $m = 0.80736 \times 20 = 16.15 \approx 16$ g Hence, the mass of gas remains constant during the heating process.
8(c)(ii) 2.	Using $Q = mc\Delta T$, $c = \frac{Q}{m\Delta T}$ $= \frac{1220}{16.14 \times \left(174 - 25\right)}$ $= 0.507$ J g^{-1} K^{-1}
8(c)(iii)	When heating at constant pressure, the gas expands. Hence work is done by the gas against the atmosphere, i.e. negative work done on gas, $-W$. From the first law of thermodynamics, $\Delta U = Q + \left(-W\right), Q = \Delta U + W$. More thermal energy is therefore required to raise the temperature of unit mass of the gas by 1 K, because some of the energy supplied is used to do work against the atmosphere. Hence, the specific heat capacity calculated **(ii)** would be larger.
9(a)	The root-mean-square (r.m.s.) value of an alternating current is the equivalent value